

Problem

A system is formed by cascading two systems as shown in Fig. 16.106. Given that the impulse responses of the systems are

$$h_1(t) = 21e^{-t}u(t), \quad h_2(t) = e^{-4t}u(t)$$

(a) Obtain the impulse response of the overall system.

(b) Check if the overall system is stable.

Figure 16.106:



Step-by-step solution

Step 1 of 2

(a)

Cascading of two systems with impulse responses represents convolution of time domain.

Convolution in time domain is equal to the multiplication of s domain.

Impulse response of the overall system is

$$\begin{aligned}
 h(t) &= h_1(t) * h_2(t) \\
 H(s) &= H_1(s) \times H_2(s) \\
 &= L(21e^{-t}u(t)) \times L(e^{-4t}u(t)) \\
 &= \left(\frac{21}{s+1} \right) \times \left(\frac{1}{s+4} \right) \\
 &= \left(\frac{21}{3} \right) \left(\frac{1}{s+1} \right) + \left(\frac{-21}{3} \right) \left(\frac{1}{s+4} \right)
 \end{aligned}$$

Applying inverse Laplace transform we get

$$\begin{aligned}
 h(t) &= \left(\frac{21}{3}e^{-t} - \frac{21}{3}e^{-4t} \right) u(t) \\
 &= (7e^{-t} - 7e^{-4t})u(t) \\
 &= 7(e^{-t} - e^{-4t})u(t)
 \end{aligned}$$

Hence the impulse response of the overall system is $\boxed{7(e^{-t} - e^{-4t})u(t)}$.

Step 2 of 2

(b)

Overall system impulse response is $h(t) = \left(\frac{21}{3} e^{-t} - \frac{21}{3} e^{-4t} \right) u(t)$

A system is said to be stable when $t \rightarrow \infty$ we get magnitude of infinite.

Put $t \rightarrow \infty$ in $h(t) = \left(\frac{21}{3} e^{-t} - \frac{21}{3} e^{-4t} \right) u(t)$ we get

$$\begin{aligned} \lim_{t \rightarrow \infty} h(t) &= \left(\frac{21}{3} e^{-\infty} - \frac{21}{3} e^{-4\infty} \right) (1) \\ &= \frac{21}{3} - \frac{21}{3} \\ &= 0 \end{aligned}$$

Hence the overall function is stable.